Some Continuous Distributions

Name	Notation	p.d.f.	Range	m.g.f.	Mean	Variance
Uniform	$\mathrm{U}(a,b)$	$\frac{1}{b-a}$	$a \le x \le b$	$\frac{e^{bt}-e^{at}}{(b-a)t}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	$\mathrm{N}(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$	$-\infty < x < \infty$	$e^{\mu t+\frac{1}{2}\sigma^2 t^2}$	μ	σ^2
Log Normal	$\mathrm{LogN}(\mu,\sigma^2)$	$\frac{1}{x\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(\ln x-\mu)^2}{2\sigma^2}\right\}$	$0 < x < \infty$	undefined for $t > 0$	$e^{\mu+\frac{1}{2}\sigma^2}$	$(e^{\sigma^2}-1)(e^{2\mu+\sigma^2})$
Gamma	$\mathrm{Gamma}(\alpha,\lambda)$	$\frac{\lambda^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\lambda x}$	$0<x<\infty$, $\alpha,\lambda>0$	$\left(\frac{\lambda}{\lambda-t}\right)^\alpha$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$
Exponential	$\mathrm{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$0<x<\infty$, $\lambda>0$	$\frac{\lambda}{\lambda-t}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Chi-square	χ^2_ν	$\frac{1}{2^{\nu/2}\Gamma(\nu/2)}x^{\nu/2-1}e^{-x/2}$	$0<x<\infty$, $\nu>0$ integer	$(1-2t)^{-\nu/2}$	ν	2ν
Beta	$\mathrm{Beta}(\alpha,\beta)$	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}$	$0<x<1$, $\alpha,\beta>0$	*omitted	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Cauchy	$\mathrm{Cauchy}(\mu,\sigma)$	$\frac{\sigma}{\pi(\sigma^2+(x-\mu)^2)}$	$\sigma > 0$	does not exist		
Pareto	$\mathrm{Pareto}(k,\theta)$	$\frac{k}{\theta}\left(1+\frac{x}{\theta}\right)^{-(k+1)}$	$x>0$, $\theta,k>0$	does not exist	$\frac{\theta}{k-1}$	$\frac{\theta^2 k}{(k-2)(k-1)^2}$
Weibull	$\mathrm{Weibull}(\nu,\lambda)$	$\nu\lambda x^{\nu-1}\exp(-\lambda x^\nu)$	$x>0$, $\nu,\lambda>0$	*omitted	$\lambda^{-\frac{1}{\nu}}\Gamma(1+\frac{1}{\nu})$	*omitted
Multivariate Normal	$\mathrm{N}_p(\boldsymbol{\mu},\boldsymbol{\Sigma})$	$\frac{\exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right\}}{(2\pi)^{p/2}\det(\boldsymbol{\Sigma})^{1/2}}$	$\mathbf{x}\in\mathbb{R}^p$	$e^{\boldsymbol{\mu}^{\mathrm{T}}\mathbf{t}+\frac{1}{2}\mathbf{t}^{\mathrm{T}}\boldsymbol{\Sigma}\mathbf{t}}$	$\boldsymbol{\mu}$	covariance matrix $\boldsymbol{\Sigma}$

Some Discrete Distributions

Name	Notation	p.m.f.	Range	m.g.f.	Mean	Variance
Discrete Uniform	$\mathrm{DU}(a,b)$	$\frac{1}{b-a+1}$	$x = a,a+1,\ldots,b-1,b$ a,b integers	$\frac{e^{at}-e^{(b+1)t}}{(b-a+1)(1-e^t)}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2-1}{12}$
Binomial	$\mathrm{Bin}(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	$x = 0,1,2,\ldots,n$ $0<p<1$	$[(1-p)+pe^t]^n$	np	$np(1-p)$
Poisson	$\mathrm{Poi}(\lambda)$	$\frac{e^{-\lambda}\lambda^x}{x!}$	$x = 0,1,2,\ldots$ $\lambda>0$ real	$e^{\lambda(e^t-1)}$	λ	λ
Hyper-geometric	$\mathrm{HypGeo}(n,K,N)$	$\frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$	$x = 0,1,2,\ldots,n$ N,K,n integers	*omitted	$\frac{nK}{N}$	$\frac{nK(N-K)(N-n)}{(N)^2(N-1)}$
Geometric	$\mathrm{Geo}(p)$	$p(1-p)^{x-1}$	$x = 1,2,\ldots$ $0<p<1$	$p[e^{-t}-(1-p)]^{-1}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Negative Binomial	$\mathrm{NegBin}(k,p)$	$\binom{x-1}{k-1}p^k(1-p)^{x-k}$	$x = k,k+1,\ldots$ $0<p<1$	$p^k[e^{-t}-(1-p)]^{-k}$	$\frac{k}{p}$	$\frac{k(1-p)}{p^2}$

*omitted due to lack of space

Empirical Bayesian Credibility theory Model 1 (Buhlmann Model)

Credibility factor

$$Z = \frac{n}{n + \frac{\widehat{\mathbf{E}[s^2(\theta_i)]}}{\widehat{\mathbf{Var}[m(\theta_i)]}}}$$

with the estimates

$$\widehat{\mathbf{E}[m(\theta_i)]} = \frac{1}{N}\sum_{i=1}^{N} \bar{X}_i,$$

$$\widehat{\mathbf{E}[s^2(\theta_i)]} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{1}{n-1}\sum_{j=1}^{n}\left(X_{ij} - \bar{X}_i\right)^2\right],$$

and

$$\widehat{\mathbf{Var}[m(\theta_i)]} = \frac{1}{N-1}\sum_{i=1}^{N}\left(\bar{X}_i - \bar{X}\right)^2 - \frac{1}{nN}\sum_{i=1}^{N}\left[\frac{1}{n-1}\sum_{j=1}^{n}\left(X_{ij} - \bar{X}_i\right)^2\right].$$